Deck

Memo

Sigma Genetics (YC W17) is pioneering the use of magnetic fields as a platform for delivering a wide range of genetic therapies, starting with CAR-T to help blood cancer patients.

We backed them in their seed round, and we're excited to follow on now with a seed extension while they seek their Series A in parallel. We are getting in at a modest step up from the prior round ($9M post-money cap SAFE → $15M post-money cap SAFE), while the Series A is targeted to be a $10M round at $40M post-money.

If you're wondering why they didn't need to raise for 4 years after their seed round, it's because they were awarded $1.75M in non-dilutive funding from Andy HIll CARE Fund and AFWERX.

Note: the Series A round already has $2.5M in soft circles from investors at the higher valuation, and as soon as Sigma finds a lead investor for that round, this SAFE will no longer be available to us.

Why We Like It

to include solid tumors. PLVs have an initial TAM of $60B, and Sigma now projects magnetoporation to have a $120B TAM by 2030.

- Their PLV-base in vivo treatments would cost 100X less than current CAR-T ex vivo treatments.
- Hit many new milestones listed below including significant efficacy and tox data, and secured a large number of partnerships that validate their platform strategy to in-license clinically validated therapies and improve them (SEE BELOW).
- Led by a team of serial entrepreneurs and PhDs.
- **Backed by Y Combinator, Westcott Investment Group, BioSphere Investment Group, Seattle Children's Research Institute, Ikarian Capital, Tacoma Venture Fund, Healthspan Capital, Andy Hill CARE Fund, IKJ Capital, Infinita, Invariantes Fund, AFWERX,** and a number of notable angels (Adam Greenberg (Iunu), Matthew Putman (Nanotronics), David Larkin (Nanotronics), Hisham Sherif).

The pipeline is robust now.

Let's go into detail on what they have achieved to date.

First, some definitions are in order:

- **CAR-T** (Chimeric Antigen Receptor T-cell): These are immune cells (T-cells) that have been genetically modified to recognize a specific antigen on the surface of cancer cells. There are FDA-approved CAR-T treatments for blood cancers, where the patient's blood needs to be treated outside the body (ex vivo). The next frontier for CAR-T includes en vivo treatments, as well as expansion to treating other cancer types. Which is where PLVs come in.
- **Magnetoporation**: Using magnetic fields to open up cells – We believe Sigma is the only company doing this, and it's far superior to electroporation for cell survival. See prior seed memo below for more details.
- **PLV** (Plasmid Lentiviral Vector): This is the delivery vehicle (vector) used to transfer the genetic code for the CAR receptor into the T-cells. PLVs were only invented six years ago, and are more effective than older AAV and LNP at stably integrating the CAR gene into the T-cell genome,

Here's why PLVs could be a big deal:

In Vivo Gene Delivery is the Bottleneck in Genetic Medicine

Feature	PLV	LNP (Lipid Nanoparticle)	AAV (Viral Vector)
Delivery of DNA	✓	✗	✓
Delivery of RNA	✓	✓	✗
› 4 kb cargo	✓	✓	✗
Extrahepatic delivery	✓	✗	✗
Safety & immunogenicity	✓	✗	✗
Repeat dosing	✓	✓	✗
Low COGS	✓	✓	✗

The Delivery Gap

→ 7 approved AAV therapies but toxicity and patient deaths remain a hurdle

→ 5 approved LNP siRNA therapies all limited to liver or vaccines

→ PLVs combine large cargo capacity, systemic extrahepatic delivery, non-immunogenic re-dosing, and scalable manufacturing

Milestones achieved since 2022 Seed round:

1) PLV CAR-T data (see publication in Cell (2024)).

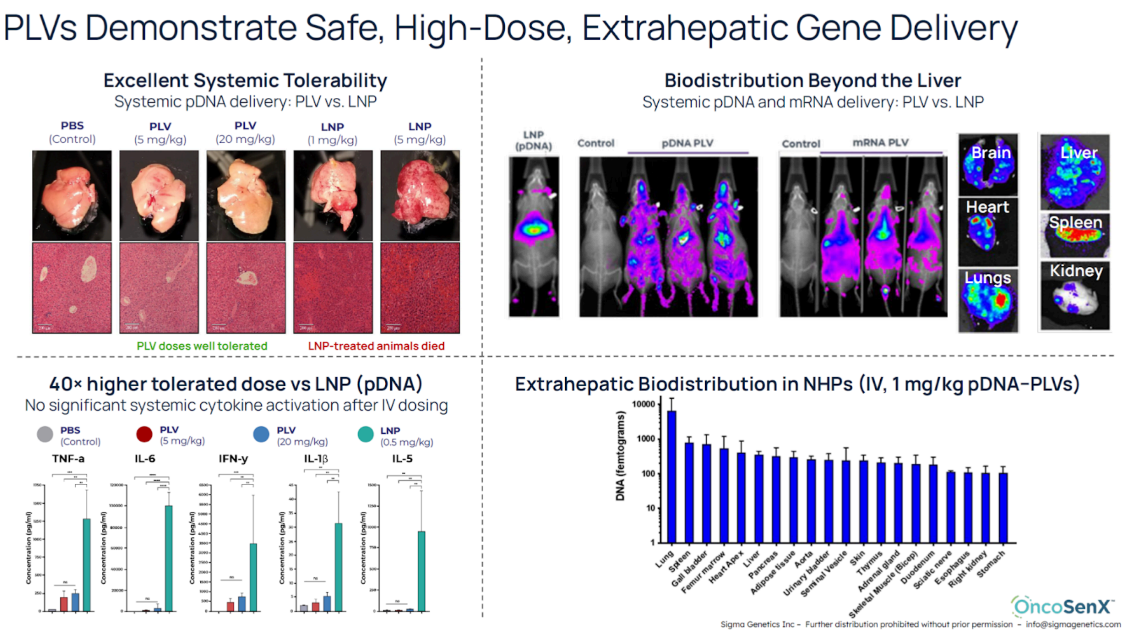
- In vivo T-cell transfection: IV administration of PLV encapsulated pDNA (200 µg) achieved selective T-cell transfection and lymphoid targeting. At day 3 post injection:
  - Lung: 33% of CD4+ and 7% of CD8+ T cells transfected
  - Spleen: 12% of CD4+ and 5% of CD8+ T cells transfected
- Functional in vivo CAR activity (anti-CD19 model): A single IV dose of therapy caused strong elimination of B cells (the target cells) in the body, showing the CAR was working effectively. By about one month, B cells were reduced to near zero in blood and spleen, while the control treatment had no effect.
- Safety profile:
  - Canine + NHP systemic tox studies: no histological changes or immune toxicity observed
  - GLP repeat dose study: n=160 rats, no treatment-related findings
  - Immunogenicity: no anti-FAST antibodies on repeat dosing in rodent, NHP, or human studies
  - Clinical exposure: safe exposure at two dose levels in 69 human subjects

2) PLV CAR-Ts in partnership with City of Hope (partner-funded)

- **Construct engineering:** Cloned each CAR-T ORF (IL13Rα2 and HER2) into our custom pDNA vector backbones (validated over years at Sigma Genetics with magnetoporation) and applied their manufacturing and vector design expertise to optimize performance and enable durable transgene expression.
- They're adding features that help the engineered T cells better find tumors, get inside them, and stay active long enough to eliminate them. Most in vivo CAR-T approaches struggle in solid

- Chemokine/trafficking modules to improve CAR-T recruitment and penetration into distal tumors (chemokine guided trafficking design). Supporting publications: CCL2 (J Immunol., 2007) and CXCR2 (Nature, 2021).
- S/MAR "expression persistence" element targeting ~2 months of transgene expression, while keeping the overall delivery approach unchanged (publication: Science, 2021).

- **Status**: First generation versions are currently being encapsulated into PLVs. Moving into in vitro screens across 12+ cancer cell lines (listed in our pitch deck) to prioritize the best constructs and PLV formulations and identify where efficacy is strongest before expanding into next validation stages.
- **IL1RAP CAR-T:** An additional JV with another City of Hope group to co-develop a first-in-class CAR-T for AML with PLVs. Sigma Genetics has an exclusive license option to PLVs for IL1RAP from OncoSenX, which is an affiliate of Entos which created PLVs (See 2024 NIH publication)



3) Magnetoporator CAR-T data

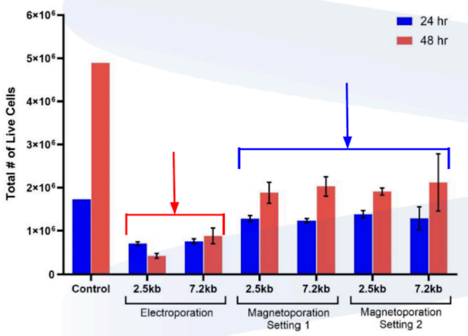
- Primary human CD4 T cells: Magnetoporation yields 16x more transfected cells and 22x more viable cells vs. electroporation at 96 hours. Transfection efficiency is 30-40% with mRNA or pDNA constructs up to 9.6 kb, with consistent performance across donors.
- CAR plasmid functionality validation: Transfected negatively selected CD4 T cells and/or pan-CD3 T cells with pDNA CAR-T constructs targeting CD22, IL13Rα2, and HER2. Across experiments, 5-10% transfection was consistently observed at 24 hours post-transfection, measured by flow cytometric detection of the surface CD34 tag/marker, confirming successful delivery and early functional expression.
- Magnetoporator commercial traction + pipeline
  - *Seattle Children's Therapeutics* (CAR-T enablement): Ongoing platform enablement work within their ecosystem (including the broader Seattle network, with ScaleReady and Andy Hill

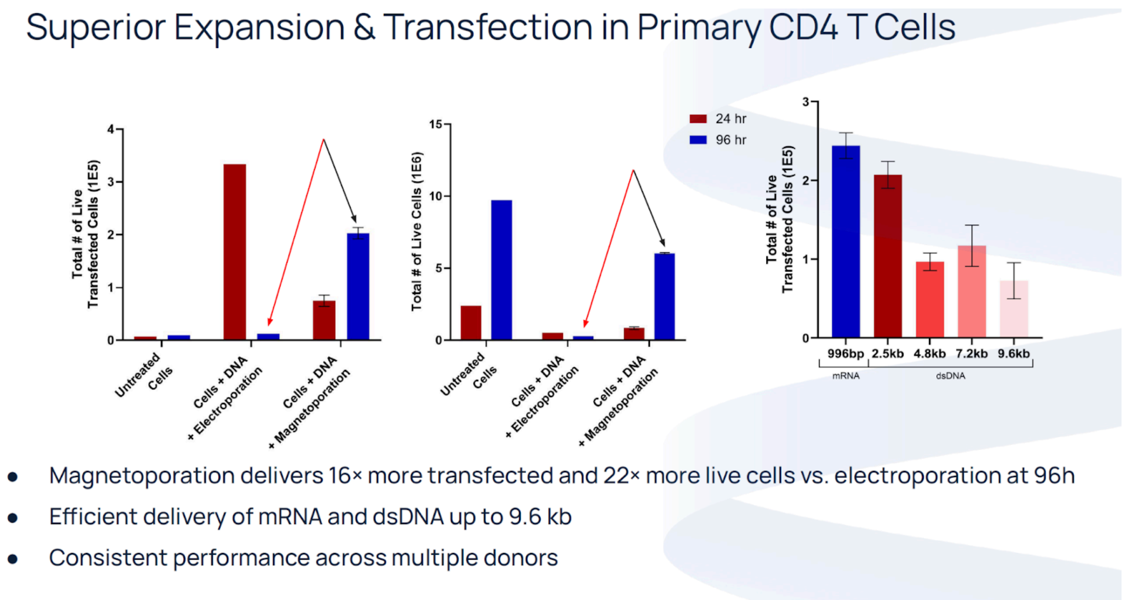
- *UCSF* (customer/collaborator: Dr. Shibani Pati, Director of the Center for Research in Transfusion Medicine and Cell Therapies): MSC-derived TIMP3 EV/exosome program for TBI. The preclinical work is DoD funded. UCSF is collaborating with Sigma to (1) design the plasmid (in build now) and (2) transfect MSCs via Magnetoporator to generate engineered EVs expressing TIMP3 for treating TBI and neurotrauma. We see 30-40% transfection in MSCs with pDNA.
- ● *Vilya Therapeutics* (customer): Vilya develops de novo macrocycles using computational and AI-driven design. Sigma is magnetoporating their AI discovered compounds into cancer cell lines to validate therapeutic signaling and function.
- ● *Accelerated Biosciences* (50/50 JV): Building an hTSC based producer cell line platform (magnetoporation enabled) for next generation biomanufacturing. For the hTSC producer cell line program, we have demonstrated up to ~40% pDNA-eGFP transfection efficiency with minimal cell loss. The incoming CRISPR knock-in construct is comparable in size to the 7.2 kb and 9.6 kb pDNA constructs we have already transfected into hTSCs. The construct includes a landing pad to insert customer sequences of interest to manufacture biologics, viral vectors, or vaccines.
- ● *Dept. of Defense / AFWERX:* Lyophilized RBC program. Sigma completed an AFWERX Phase I SBIR, validating that they can load RBCs with cryoprotectants via magnetoporation, and have been invited to apply for a Phase II SBIR. The Phase II SOW funds R&D to finalize a manufacturing SOP. Submission is pending Congress reopening the SBIR pool to formally submit the grant.
- ● *ARPA-H* (BoSS submission): Sigma is part of an ARPA-H BoSS (Technical Area 2: Bioprocessing Capabilities) submission in collaboration with UT Austin (Dr. Zhengrong Cui) and Via Therapeutics. The program focuses on scalable bioprocessing and thin-film freeze drying approaches to enable stabilized, rehydratable formulations for sensitive biologics and cell-based products. Sigma contributes magnetoporation-enabled cell engineering capabilities as part of the integrated manufacturing solution, in exchange for a $300k allocation in the proposal.

Magnetoporation vs. Electroporation Cell Viability

Magnetoporation vs. Electroporation using varying DNA construct sizes.

- Gene inserts of 2.5kb and 7.2kb were tested. The number of magnetoporation transfected GFP+ cells increased from 24 to 48 hours relative to the number of electroporation transfected GFP+ cells which resulted in substantial cell death.
- Magnetoporated CD4 T cells demonstrated greater proliferative capacity and survival vs. electroporated samples.





AFWERX Phase I SBIR to investigate the ability to use magnetoporation to effectively freeze blood for long-term use.

- Title: Advancing battlefield and global blood supply solutions by novel magnetoporation: Superior trehalose delivery to create functional lyophilized red blood cells
- Demonstrated magnetoporation enabled intracellular trehalose delivery (>100 mM) into human RBCs with low hemolysis, establishing TRL-3 proof of concept for scalable lyophilized RBC manufacturing
- Secured USAF 59th Medical Wing (DAF) / DOW customer; aligned with battlefield transfusion needs and procurement interest for 50,000 units
- Invited to AFWERX Phase II Application
  - Scope includes continuous-flow magnetoporation, cryo-studies, optimized lyophilization/rehydration protocols, and in vivo transfusion validation
  - Confirmed & Engaged DoD End Users
    - USSOCOM (SOF medics, austere transfusion operations)
    - U.S. Air Force (59th Medical Wing; AFSOC; AFRC)
    - U.S. Army Institute of Surgical Research (ISR) – joint service applicability & battlefield hemorrhage care

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2022 Seed Memo excerpts:

Introduction

We believe one of the most incredible cancer therapies of this generation , and a victory for gene therapy, is undoubtedly [CAR-T therapy](). Although the potential was first understood in the 80s, the first FDA-approved treatment didn't appear [until 2017 (for Kymriah)](). In this treatment, the patient's blood is removed, the T-cells are genetically modified to fight cancer, and then the blood is transfused back into the patient.

AAV and LNP pose various challenges, and there are a wave of startups now exploring alternative vectors for safely and accurately targeting the delivery of genetic material to cells.

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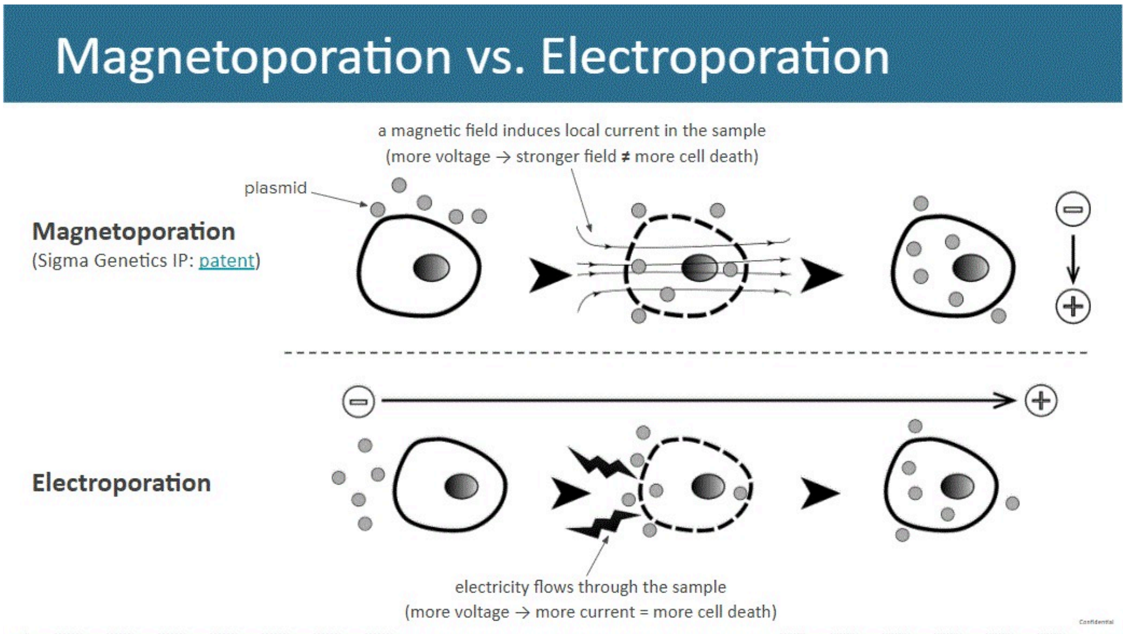
Intracellular gene delivery is the Achilles Heel of gene therapy

Although there are many incredible genetic therapies on the horizon, getting mRNA and other genetic material into the target cells remains a massive challenge. The most conventional methods such as AAV and LNP are expensive, and often involve tradeoffs between targeted accuracy of delivery (i.e. efficacy) vs. causing cell death or immune system response (i.e. side effects).

In the case of CAR-T cancer therapy with conventional FDA-approved vectors, downstream per patient costs can exceed $2.5M, because those therapies are manufactured using viral transduction that can cost up to $30k per dose.

Sigma's Magnetoporation aims to enable safe and consistent delivery - decreasing CAR-T costs, speeding treatment and helping more cancer patients

Sigma uses magnetic fields to open cell boundaries without damaging the cell (not to be confused with electroporation alternatives):



We believe this is a more scalable solution than AAV or LNP, and doesn't require making tradeoffs between cell survival, immunogenicity and effective targeted delivery. It is also faster, likely cutting total treatment time in half from a week to 2-3 days. Sigma's device has been demonstrated in vitro to operate at levels of <1% cell death (far lower than viral transduction or electroporation). It provides rapid and instant transfection of the cells for gene delivery, scalable to > 1 billion cells per reaction, and has lower regulatory requirements than conventional alternatives.

	Viral	Electroporation	Microfluidics	Magnetoporation
Cell Survival	✓		✓	✓
Consistent Delivery	✓			✓
Non-immunogenic		✓	✓	✓
Scalable			?	✓
Companies	NOVARTIS, GILEAD, OxfordBioMedica, Bristol-Myers Squibb, bluebirdbio	inovio, ONCOSEC, MaxCyte, Miltenyi Biotec, ThermoFisher SCIENTIFIC, Lonza, BTX, BIO-RAD	SQZBIOTECH, indee labs, CellFE, KYTOPEN	SigmaGenetics

Sigma has already patented and developed their device, and secured important partnerships with Eagle Harbor Technologies for manufacturing and [Seattle Children's Research Institute for research] that will lead to FDA 510(k) approvals. Note that one of Seattle Children's key executives, Dr. Michael Jensen, was a founder of Juno Therapeutics, a leader in CAR-T therapy acquired by Celgene for $9B.

Sigma's "printer/ink" business model is robust and profitable, and yields growing revenues per customer as FDA trials progress

After discussions with dozens of potential customers, Sigma believes it can price its device at $30K for research institutions, and $50k for pre-clinical large pharma. For GMP systems used in clinical trials, $100K per device + pricing evolves to a per-patient dose model, scaling from $5,000 per patient dose in clinical trials to $7,500 per patient dose for FDA-approved commercial drugs.

Sigma's team has solid science and domain expertise (and a lot of prior startup experience)

- Zachary Rapp (CEO / co-founder): co-founder, **PhenoMX**; Founding Team and Headof Business Development, Consultant at **Pharmagellan**, **Altoida**; Senior Associate Researcher (cancer), Icahn School of Medicine at **Mount Sinai**; Masters degree in Biotechnology, **Columbia University**.
- Chris Pihl (Board of directors / co-founder): CTO/co-founder at **Helion Energy**; President, **Pulse Power Solutions**; Engineer / Scientist II, **Plasma Dynamics Lab, University of Washington**; Pulsed Power Engineer, **MSNW**; BA, Innovation and Technology Management, **University of Washington**.
- Matt Scholz (Board of directors / cofounder): CEO/founder, **Oison Biotechnologies**; CEO, **OncoSenX**; CEO/co-founder, **Aegis Life**; CTO/founder, **Immusoft Corp**; CEO, **Point B Telematics**; B.S., Computer Science, **University of Washington**.
- Dr. Heather-Marie Wilson, PhD (Senior Scientist, Cell Biology and Transfection): Volunteer Staff Scientist, **UW Institute for Stem Cell and Regenerative Medicine**; Staff Scientist, Hope Heart Matrix Biology Program, **Benaroya Research Institute;** Senior Fellow, Dept of Radiation Oncology, **University of Washington;** Ph.D, Pathology **University of Washington School of Medicine.**
- Dr. Melissa Jackson, PhD (Scientist): Development Manager, **Sony Biotechnology**; Research Scientist III / Lab Manager, **Pacific Northwest Diabetes Research Institute**; Ph.D, Genetics and

- James Overaa (Senior Electrical Engineer)

Co-investors

WG Westcott Investment Group
Undisclosed

IC IKJ Capital
Undisclosed

Invariantes Fund
Undisclosed

All documents

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Document	Uploaded
Sigma Genetics Seed Pref. Deck	Feb 25, 2026
Closing documents	Feb 27, 2026

Past financing

Seed in 2022 from YC, Ikarian Capital, Seattle Children's Hospital, Westcott, Biosphere, and various angels.

Risks and disclaimers